



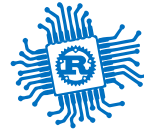
Memory Protection

Lecture 9



Memory Protection

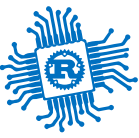
- Memory Protection Unit
- Memory Management Unit



Memory Protection

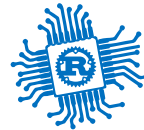
ARM: MPU, RISC-V: PMP





Memory Management

MMU



Bibliography

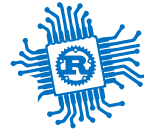
for this section

1. **Andrew Tanenbaum**, *Modern Operating Systems (4th edition)*

- Chapter 3 - *Memory Management*
 - Subchapter 3.3 - *Virtual Memory*

2. **Philipp Oppermann**, *Writing an OS in Rust*

- *Introduction to Paging*
- *Paging Implementation*



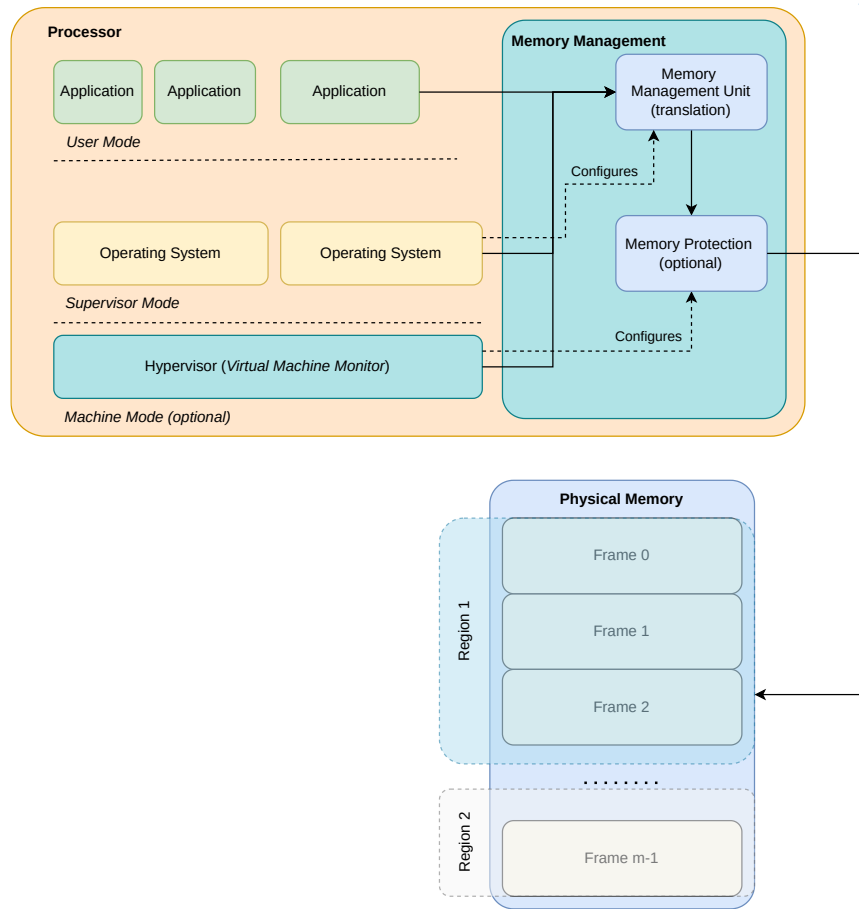
Memory Management

memory access defined page by page

- uses *logical addresses*
- **translates** to *physical addresses*

The processor works in at least two modes:

- **supervisor mode**
 - restricts access to some registers
 - accesses virtual addresses through Memory Protection (*if machine mode exists*)
- **user mode**
 - allows only ALU and memory load and store
 - accesses memory access through the Memory Management Unit (*MMU*)





Paging

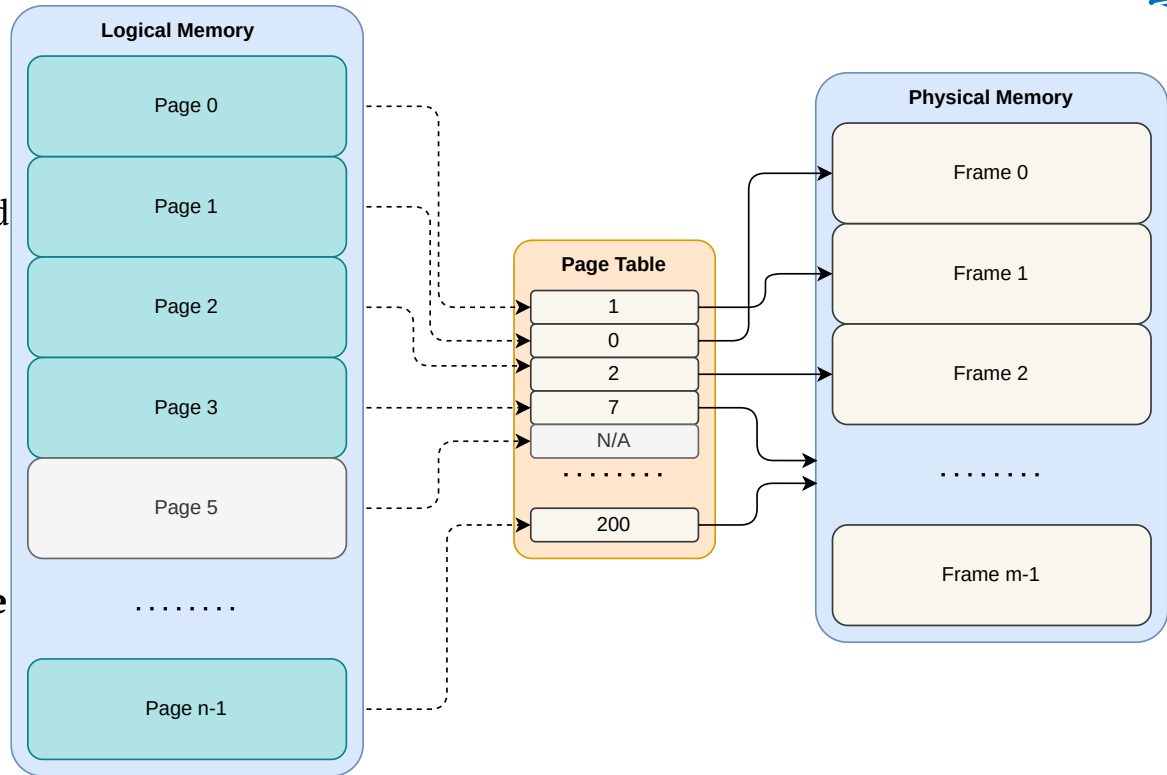
the memory *unit* is the page

- Physical Memory (*RAM*) is divided in **frames**
- Logical Memory is divided in **pages**
- $page = frame = 4 \text{ KB}$ (usually)

logical addresses are translated to *physical addresses* using a **page table**

the **page table** is located in the **physical memory**

- each memory access requires at least 2 memory accesses





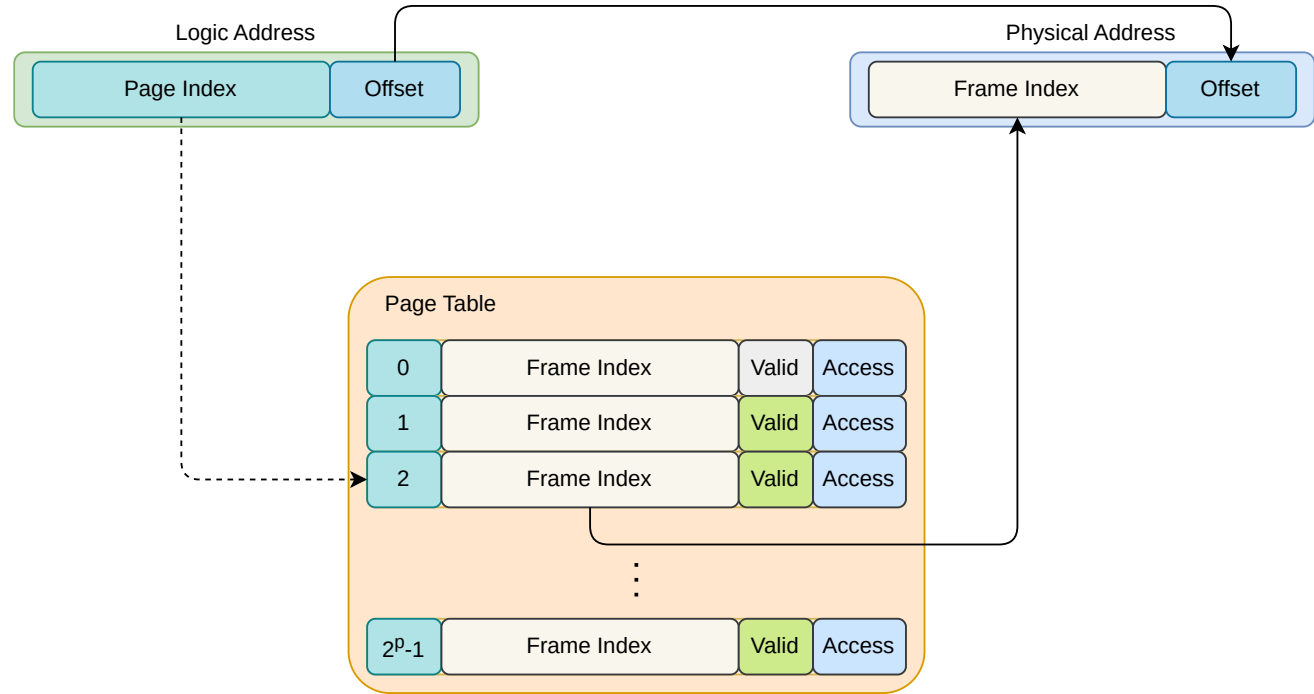
Address Translation

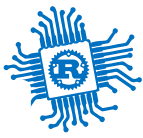
page to frame

the logic address is divided in two parts:

- *page index*
- *offset* within the page

the MMU translates every logic address into a physical address using a *page table*





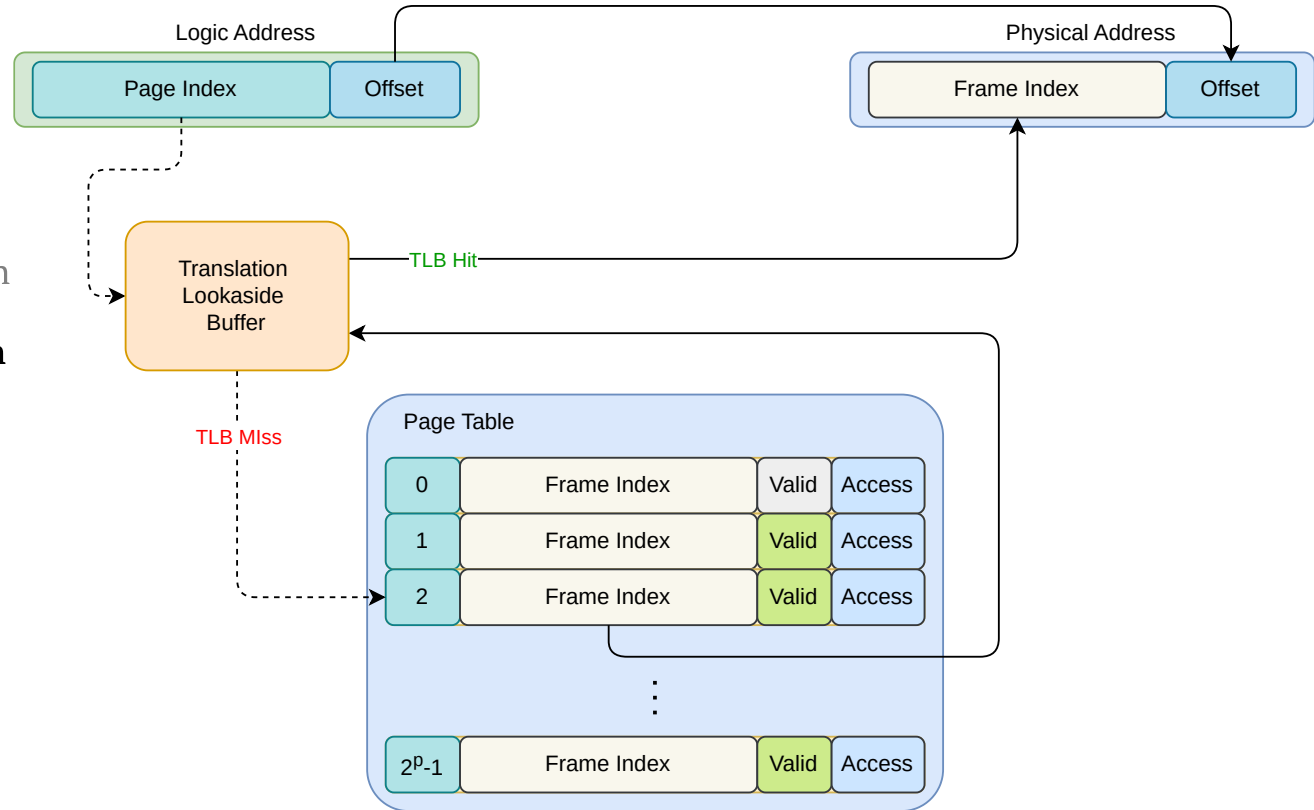
Translation Lookaside Buffer (TLB)

caching address translation

the **page table** is stored in **RAM**

each memory access **requires 2 accesses**

1. read the page table entry to translate the address
2. the requested access





Page Directory

caching address translation

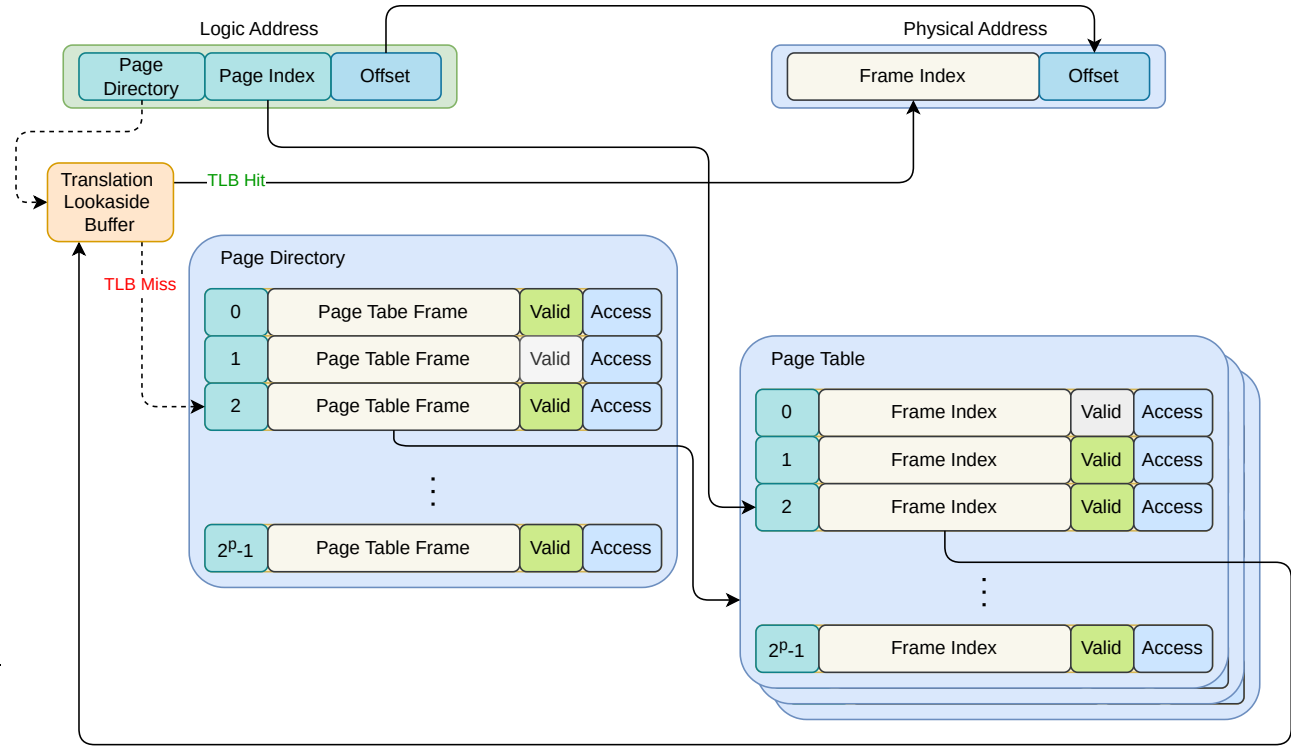
$$size_{table} = \frac{size_{ram}}{size_{page}}$$

- each table entry is 4B
- the address space is 4GB
(for 32 bits processors)

$$size_{table_32_bits} = \frac{2^{32}}{4 \times 2^{10}}$$

$$size_{table_32_bits} = 4MB$$

RAM was counted in MB
when paging started being
used



two levels, page directory and table, usually used for 32 bits systems

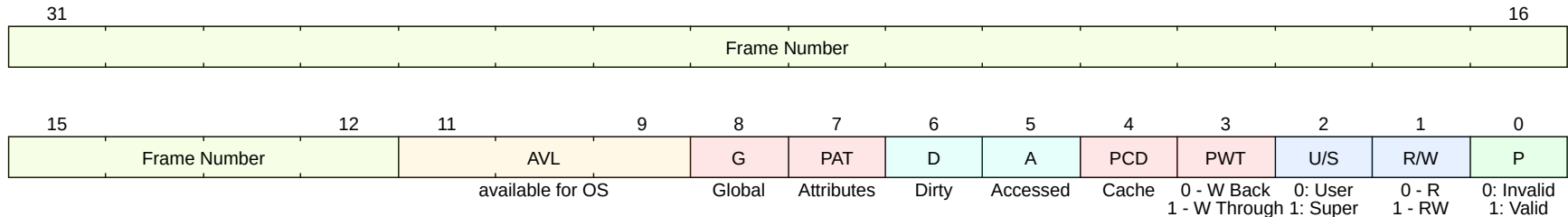


Page Table Entry

for x86 - 32 bits

this is one entry of the page table

- **P** - is the page's frame present in RAM?
- **R/W** - read only or read write access
- **U/S** - can the page be accessed in user mode?
- **D** and **A** - has this page been written since the OS has reset these bits?
- **AVL** - bits available for the OS to use, ignored by MMU



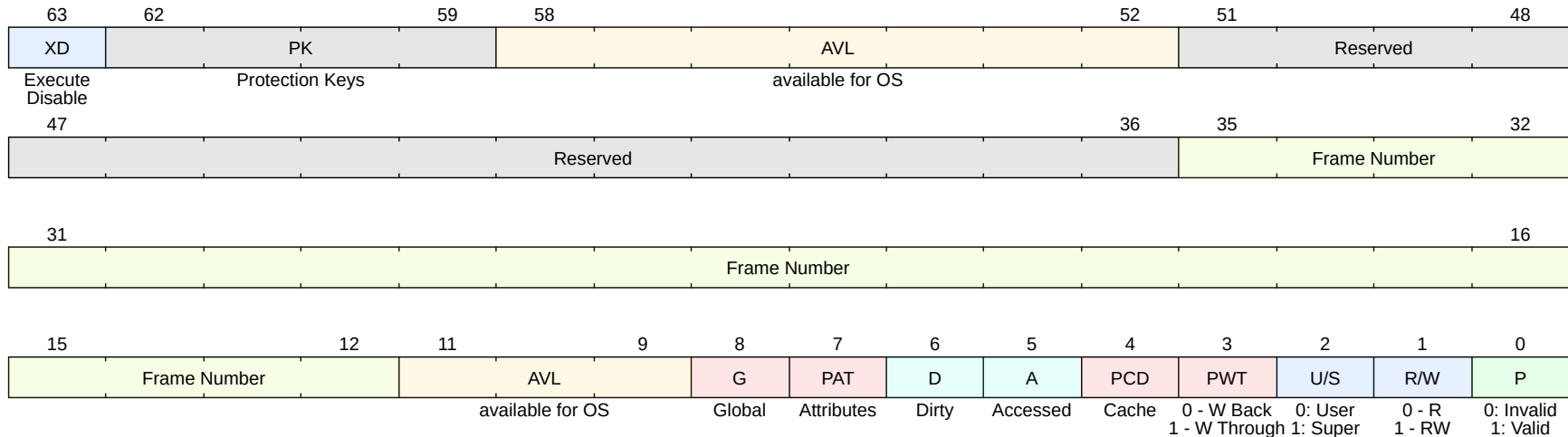


Page Table Entry

for x86 - 32 bits with PAE

this is one entry of the page table using Physical Address Extension (*PAE*)

- **XD** - eXecute Disable (aka *DEP*), if set triggers a fault if an instruction is read from the page
- **PK** - Protection Keys, allows user mode to set protection (64 bit only)

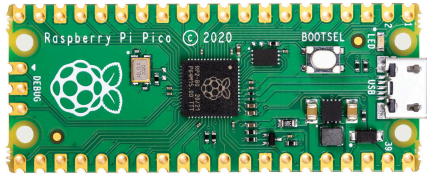




Microcontroller (MCU)

Integrated in embedded systems for certain tasks

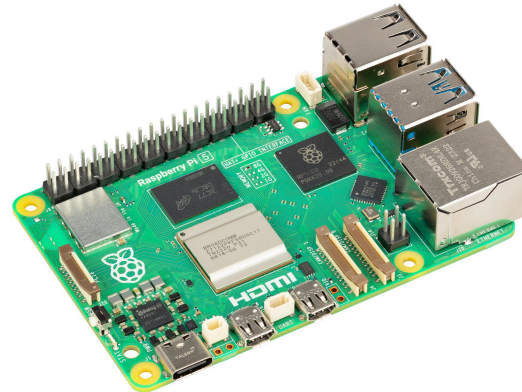
- low operating frequency (MHz)
- a lot of I/O ports
- controls hardware
- does not require an Operating System
- costs \$0.1 - \$25
- uses **Memory Protection Unit**

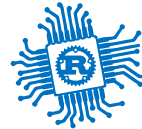


Microprocessor (CPU)

General purpose, for PC & workstations

- high operating frequency (GHz)
- limited number of I/O ports
- usually requires an Operating System
- costs \$75 - \$500
- uses **Memory Management Unit**





Conclusion

we talked about

- Memory Protection Unit
- Memory Management Unit